troubleshooting, input buffers for speedier printing, the manuals and documentation provided, along with other accessories such as a bundled data cable, specialised paper and software. We also checked for miscellaneous features such as support for dual-sided printing, printing on iron-on transfers, photo printing support and transparency printing.

Performance: We evaluated performance on two basic criteria:
a) Printing speed: We used a sample document file with 25 per cent text coverage and logged the time taken to print this in black and white mode. We used ordinary paper and the print mode was set at normal. We used a colour photograph for the colour speed test. We logged the time taken to print this in best quality mode on coated photo-quality inkjet paper and to print it in greyscale mode (normal print quality mode) on ordinary paper.
b) Print quality: The first document we used to test the print quality was a combination of text, vector graphics and photos. We observed specific portions of this combi document to evaluate certain characteristics of the printer (see image below). The second print was of a photograph, the same one used for the speed test, which was used to evaluate the ability of the printer to handle variations in colour, hue and brightness. It also had many fine details and specular effects that would bring out the printing capability of the printer. This full colour photograph was printed in best quality mode, on a coated inkjet paper.

To test the quality of text printing, we used a text printout which was divided into four zones. Each zone was observed for flaws such as smudging, smearing, jaggedness, misaligned text, faded text, crispness and depth of blackness. Each zone was also examined for repeatability of the errors and points were then awarded suitably.

1: Uniformity of the outline circle: We checked for misalignments and missing patches.
2: Resolution test: We examined the printer's ability to reproduce closely packed converging lines.
3 and 4: Quality of photos: Specific details for each photo was mapped (clarity of the grille pattern, the shine on the fruit and so on).
5: Yellow text on black background: This area helped us evaluate the printer's ability to print high-contrast text.
6: Number of distinct colour bars: There are 10 bars in each colour pattern and each bar has two shades. If a printer could print two distinct shades for each bar in every colour pattern, it was awarded a maximum score of 10.
7: Readability of fine text: This text is in varying point sizes ranging from 6 to 2 points. The printer was awarded one point for each readable line.

Cost of ownership (cost per page): Inkjets drink up ink really fast. For instance, a cartridge that costs you Rs 1,000 could end up having a capacity of not more than a couple of hundred pages. Hence, over a year you could run up quite a huge bill in just replenishing the ink cartridges. This is why the cost of ownership should be an important consideration when buying an inkjet printer.

We analysed the long-term costs associated with printers by running a special cartridge drain test. This test evaluates the cost per page for printing colour graphics as well as for printing black text. We first installed a new ink cartridge and printed two documents over and over again till all the ink was used up, or till the printer refused to continue. The first document was created in Photoshop and was composed of alternating groups of cyan, magenta and yellow bars with no blank spaces in between. The entire print area was covered by these bars, with each bar covering 33 per cent of the print area. This was the CMY drain test. The second document had 90 per cent pure black bars and 10 per cent blank space. This was the drain test for the black (K) cartridge.

The cost of ownership test yields two cost-per-page numbers, one for the text-based documents (the black ink cartridge) and one for graphics-heavy prints (the colour ink cartridge).

But most documents that people print, such as Web pages, text manuals, colour photos, etc, are usually a mix of black text and colour images. The average ink coverage on paper is only 16 per cent (8 per cent black and 8 per cent colour). However, to print colour (CMYK) you also need to add black ink, therefore 25 per cent black ink is used in every colour printout as well! The weightages actually turn out to be 66 per cent for black (10 per cent coverage) and 33 per cent for colour (6 per cent coverage). Hence the exact calculation is:

Black = 8 per cent + (one-fourth of 8 per cent of colour).

We applied this formula to the results of our cartridge drain test to come up with the cost per page in Rs (factoring in the price per replacement cartridge).

Warranty and support: Here we took into account the warranty period for the inkjets, the number of authorised service centres appointed by the company and the number of cities in which authorised service centres are present.

Value for money: This is a factor of the performance and features offered by the printer as compared to its price. Therefore, the better the performance and features and the lower the price, the higher will be the value for money of a printer. We computed this value by dividing the sum of an inkjet's scores in performance, features and the total cost of ownership by its price to obtain a value for money index.

Test Analysis

Inkjet printers have shown a marked improvement in not just their core ability to print, but also in delivering the most bang for the buck in terms of running costs and features. Here is how